



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Name: Abobakr Mohamed

PRN: 25070122015

Branch: CSE (2025-29)

Semester: 3

Subject: Programming with Java

ASSIGNMENT No.: 4

TITLE: :Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement:

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives:

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory:

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Outcomes:

Wrapper classes in Java (Integer, Double, Character, Boolean, etc.) allow primitive data types to be treated as objects. This experiment demonstrated how to convert primitive values into their corresponding wrapper objects (boxing) and back into primitives (unboxing), how to convert String input into usable numeric types using methods like `parseInt()` and `parseDouble()`, and how wrapper class methods can be used to perform comparisons and basic operations on wrapped values — concepts essential for working with Java Collections and handling real-world user input.

Conclusion:

This experiment demonstrated the use of wrapper classes to convert between primitive data types and objects. Methods like `parseInt()` and `parseDouble()` converted String input into usable numbers, while `compareTo()` enabled validation, showing how wrapper classes support real-world data handling in Java.

Exercise:

1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
class Student {  
  
    String name;  
    String mark1Str, mark2Str, mark3Str;  
  
    Student(String n, String m1, String m2, String m3) {  
        name = n;  
        mark1Str = m1;  
        mark2Str = m2;  
        mark3Str = m3;  
    }  
  
    void calculateTotal() {  
        int mark1 = Integer.parseInt(mark1Str);  
        int mark2 = Integer.parseInt(mark2Str);  
        int mark3 = Integer.parseInt(mark3Str);  
  
        int total = mark1 + mark2 + mark3;  
        double average = total / 3.0;  
  
        System.out.println("Name: " + name);  
        System.out.println("Marks: " + mark1 + ", " + mark2 + ", " + mark3);  
        System.out.println("Total: " + total);  
    }  
}
```

```
        System.out.println("Average: " + average);
    }
}

public class JAVA_EXP_3 {

    public static void main(String[] args) {

        Student s1 = new Student("bkor", "75", "80", "90");
        s1.calculateTotal();

    }
}
```

```
[Running] cd "c:\Users\M-Mamoun Albrair\OneDrive\Desktop\New folder (4)\\" && javac JAVA_EXP_3.java && java JAVA_EXP_3
Name: bkor
Marks: 75, 80, 90
Total: 245
Average: 81.66666666666667

[Done] exited with code=0 in 2.988 seconds
```

2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
class Employee {

    String empId;
    Double basicSalary;
    Double bonus;

    Employee(String id, String salaryStr, String bonusStr) {
        empId = id;
        basicSalary = Double.parseDouble(salaryStr);
        bonus = Double.parseDouble(bonusStr);
    }

    void calculateNetSalary() {

        if (basicSalary.compareTo(0.0) <= 0) {
            System.out.println("Invalid basic salary for Employee ID: " + empId);
            return;
        }

        if (bonus.compareTo(0.0) < 0) {
            System.out.println("Invalid bonus amount for Employee ID: " + empId);
            return;
        }

        double netSalary = basicSalary + bonus;

        System.out.println("Employee ID: " + empId);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("Bonus: " + bonus);
        System.out.println("Net Salary: " + netSalary);
    }
}

public class JAVA_EXP_3 {

    public static void main(String[] args) {

        Employee e1 = new Employee("E015", "45000", "5000");
        e1.calculateNetSalary();

        System.out.println();

        Employee e2 = new Employee("E102", "-2000", "1000");
        e2.calculateNetSalary();

    }
}
```

```
[Running] cd "c:\Users\M-Mamoun Albrair\OneDrive\Desktop\New folder (4)\ " && javac JAVA_EXP_3.java && java JAVA_EXP_3
Employee ID: E015
Basic Salary: 45000.0
Bonus: 5000.0
Net Salary: 50000.0
|
Invalid basic salary for Employee ID: E102

[Done] exited with code=0 in 4.118 seconds
```